

Joint Error Bounds for Finite Quantum Controllers, Output Capture, and Module Renewal

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Abstract

Finite quantum control architectures require error guarantees that remain valid when clocks, reference systems, output records, and correlated external registers are retained. We analyze a prepared architecture consisting of a prefix-dressed perfect-transfer history clock, an output capture operation, and renewal by exchange with a fresh matched module. For timed capture we bound incomplete transfer and false terminal tags, and identify the factorization assumption under which the stored output follows its declared free evolution after the capture contact is switched off. A distinct static capture construction provides a finite retention interval: its complete joint-state error scales as the square root of the untransferred clock population, whereas a reduced-body bound scales as the population itself. Renewal transfers the old module’s correlations into the outgoing donor register rather than erasing them. A compositional trace-distance budget accounts for the complete resource bank and retains failure histories. We also state a stationary-input preparation obstruction and the obstruction to exact permanent first-entry storage in a closed finite static model. Exact finite-map checks and small numerical fixtures support implementation verification. The constructions assume prepared states, specified couplings, and supplied launch and routing controls; they establish neither a complete autonomous cycle nor a device-level fault-tolerance threshold.

Index Terms: Quantum control, quantum clocks, quantum memory, trace distance, coherent resources, quantum channels.

1. Introduction

Quantum computation is commonly specified as a sequence of ideal gates, but a finite control architecture also contains a clock, prepared reference systems, capture controls, output storage, and the resources needed for another execution. These registers can remain correlated with one another and with earlier outputs. Consequently, high fidelity of a reduced computational state is insufficient to certify a complete execution-and-renewal cycle. The engineering question addressed here is how to preserve a consistent comparison between a stipulated implementation model and its ideal reference when these registers and all failure histories remain in the domain.

We study three operations: executing a supplied gate word, transferring its output to a matched storage register, and exchanging a consumed active module with a fresh prepared donor. Their synthesis uses established operator constructions. Our emphasis is the scope and composition of their guarantees: which registers are retained, which controls are assumptions, which error terms overlap, and which finite resources are consumed. The framework concerns finite quantum-control and memory interfaces. It does not supply an error-correcting code or a scalable quantum-computing architecture.

1.1 Related work and contribution boundary

Engineered perfect state transfer on a spin chain is established by Christandl et al. [1]. The clock below has $n + 1$ addresses and uses their square-root coupling structure. Prefix-dressed propagation belongs to the Feynman-Kitaev history-clock framework. Caha, Landau, and Nagaj [2] analyze history clocks, idling, and pulse tuning. Their ground-state and time-averaged results should not be identified with the pointwise real-time retention estimate considered here. Woods, Silva, and Oppenheim [3] study finite clocks and their backreaction through a different clock construction. We establish no equivalence or performance advantage over it. Coherence-broadcasting limitations [4] distinguish an unchanged catalyst from a consumed coherent resource. Unitary flag premeasurement [5], perturbation estimates, and SWAP identities are also established ingredients.

The proposed contribution is a coordinated finite-domain analysis of execution, capture, and renewal. Its candidate value is the distinction between complete and reduced-state errors, the different timed and static storage guarantees, and a resource-bank admission condition that prevents treating correlated fresh slots as independent. These are contribution candidates to be assessed against prior work, not a completed novelty claim. In particular, the renewal identity alone is elementary unitary relabeling.

The mathematical development arose from earlier finite-operator studies [6]. The analysis here is stated directly on finite complex Hilbert spaces and does not require a tetrahedral spatial interpretation. Earlier source constructions [7]-[10] provide provenance, while the proofs and contracts retained here define the claims used in this draft.

1.2 Results and engineering interpretation

The history clock executes the prepared word at an ideal endpoint and returns later; return of its address does not replenish resources. A timed guarded exchange captures the terminal branch with a perturbation bound. Once the pulse is off, an isolated output register evolves freely under its stated Hamiltonian. This is an exact model statement, not a measured storage lifetime. Static capture, by contrast, retains a live clock and supplies only a finite window before ideal reversal. Matched renewal restores the active module only by consuming an independent prepared donor and retaining the old correlated state elsewhere.

The error ledger is unconditional. A failed attempt consumes its allocated resources, and a conditional accepted-state estimate must be reported together with its acceptance probability. Complete-bank initialization is charged once. Each operation must have a comparison valid on the same joint domain, including spectators. Neither the analysis nor the numerical fixtures establish a tolerable physical error rate, a logical-error threshold, or hardware energy efficiency.

1.3 Organization and evidence status

Section 2 defines the joint domain and distance. Sections 3 and 4 give coherent-resource and history-clock contracts. Sections 5-7 distinguish timed protection, closed-system obstructions, and static retention. Sections 8 and 9 treat renewal, composition, and energy conventions. Section 10 states reproducible finite examples, and Section 11 identifies the remaining engineering questions. This is an IEEE-focused research development draft; journal-template preparation, complete novelty assessment, and independent mathematical review remain pending.

2. Joint domains, distances and error ownership

All spaces in this paper are finite-dimensional complex Hilbert spaces. Tensor identities are suppressed when their domain is declared. Hamiltonians are Hermitian. We use $\hbar = 1$ as a mathematical convention; no laboratory unit or action normalization is inferred. The operator norm is $\|A\| = \sup_{\|v\|=1} \|Av\|$. For states,

$$D_{\text{tr}}(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1 \in [0, 1].$$

A joint state includes every named register and arbitrary spectator Y , possibly correlated with the active body. Any free spectator dynamics is common to actual and comparison channels. A marginal statement explicitly names the traced registers. Probabilities, state distances and Hamiltonian norms have different types.

The work module is $W = (S, R, F)$: computational work bank S , reference R and fuel F . Production clock C has $n + 1$ addresses. Capture adds a matched output bank O and degenerate flag L ; optional degenerate memory/witness registers are M, E . Static capture adds address clock Q of dimension $N + 1$. Donor D matches the complete module being exchanged. Y includes prior outputs, records, waste and every retained external correlation.

Matched means equal carrier dimensions and equal bare energy operators under the declared identification. It is stronger than equal mean energy. A blank output or fresh donor state is a supplied resource, not a state created by exchange.

Lemma 1 (joint perturbation). If H, K are Hermitian and $\|H - K\| \leq d$, then for any $t \geq 0$,

$$\|e^{-itH} - e^{-itK}\| \leq td, \quad D_{\text{tr}}(U_H \rho U_H^\dagger, U_K \rho U_K^\dagger) \leq \min\{1, td\},$$

including arbitrary spectator extensions. For a time-dependent extra contact $V(u)$ on the same domain, replace td by $\int_0^t \|V(u)\| du$.

Proof. Duhamel writes the propagator difference as an integral of unitary factors around $H - K$; submultiplicativity bounds it by td . On a purification, the vector difference has that bound, and pure-state trace distance is at most vector distance. Reduction contracts trace distance. The time-dependent proof uses the corresponding propagator identity. $\square$

If an event projector P annihilates the comparison vector, its actual probability is at most $\min\{1, (td)^2\}$ on that purification. If comparison event amplitude is at most a , the actual probability is at most $\min\{1, (a + td)^2\}$. An initial full-joint state discrepancy instead changes any event probability by at most its trace distance; it is not automatically an amplitude-square debit.

All budgets below refer to a fixed pair of actual and ideal contracts. For the unitary channels $\mathcal{U}_H$ and $\mathcal{U}_K$, the uniform spectator extension also gives

$$\frac{1}{2} \|\mathcal{U}_H - \mathcal{U}_K\|_\diamond \leq \min\{1, td\}.$$

Here the diamond norm is the completely bounded trace norm of the channel difference. This is a channel interpretation of the same perturbation estimate, not a new physical noise model.

Additional supplied-state, contact, thermal, detector, storage and implementation errors must be admitted on matching domains. Shared errors are charged once, not repeated for each marginal or each quoted diagnostic.

3. Finite reference and coherent binary fuel

3.1 Finite ladder and declared production target

Supply an energy spacing $\Delta > 0$, a packet length L , m source cells, and a certified integer shift bound C_* . A finite reference ladder may be chosen with

$$M_R = L + 2(m + 1)C_*, \quad n_0 = (m + 1)C_*,$$

$$H_R = \Delta \sum_{a=0}^{M_R-1} a|a\rangle\langle a|.$$

The packet occupies $n_0, \dots, n_0 + L - 1$. Up to $m + 1$ admitted shifts of magnitude at most C_* stay inside this ladder. Gate extensions on the remaining states must be specified; no reference wraparound is assumed.

An implemented work operator H_W and a comparison operator $\bar{H}_W$ are distinct supplied objects. A bound $\|H_W - \bar{H}_W\| \leq d$ must hold on the complete admitted domain. It is not a consequence of choosing a ladder. Source-specific cutoff and overlap certificates in [7]-[8] are examples of how such premises may be supplied, rather than properties of an arbitrary work Hamiltonian.

Write η_{bank} for a separately certified complete source/reference target error of isolated production. When a later result invokes this parameter, the certificate must specify the target, support, and allowed contacts. If another

operation acts on the same reference, fuel, or clock, an isolated-production certificate alone is insufficient: the integrated joint comparison must be supplied. Preparation error already included in complete-bank admission must not be charged again under this symbol.

3.2 Energy-preserving producer

Theorem 2 (binary fuel). Let $L = 2^q$, $\Delta > 0$, and supply independent fuel qubits F_j in $|+\rangle$, with $H_{F_j} = \Delta 2^j |1\rangle\langle 1|$. Supply reference seed $|n_0\rangle$. There are disjoint energy-preserving exchanges that send

$$|+\rangle_F^{\otimes q} |n_0\rangle_R \mapsto |0\rangle_F^{\otimes q} |\eta_L\rangle_R, \quad |\eta_L\rangle = \frac{1}{\sqrt{L}} \sum_{r=0}^{L-1} |n_0 + r\rangle.$$

The seed energy $n_0\Delta$ and all fuel coherence are inputs.

Proof. At step j pair $|1\rangle|n_0 + r\rangle$ with $|0\rangle|n_0 + r + 2^j\rangle$ whenever the j th bit of r is zero in the declared packet block. Pairs are disjoint, preserve the sum of the two bare energies, and define a unitary permutation with identity on the complement. By induction, before step j the populated reference labels are $r = 0, \dots, 2^j - 1$, so the current bit is zero. The next exchange doubles the uniform reference support and leaves fuel bit j in ground state. After q steps every binary label occurs once. $\square$

For the bit rule write $r = 2^{j+1}u + v$ and $r \equiv v \pmod{2^{j+1}}$, with $0 \leq v < 2^j$. Transfer replaces v by $v + 2^j$ on the declared pair. This rule defines exchange pairs rather than a recycling reservoir.

For ideal independent fuel, elementary geometric sums give

$$\langle H_F \rangle = \frac{\Delta(L-1)}{2}, \quad \text{Var } H_F = \frac{\Delta^2(L^2-1)}{12}.$$

For a pure state and the specified time-translation generator, $F_Q = 4 \text{Var } H_F$. Correlated pure fuel requires the variance of the complete sum, including covariance terms. For mixed fuel this equality is not generally valid. Arbitrary correlated fuel is transformed to its corresponding joint output; the producer does not replace it by an ideal independent packet.

3.3 Stationary complete-supply limitation

Theorem 3 (stationary supply). Let the complete input be stationary under a declared additive total free-energy generator. Let the complete production channel be covariant for its input/output generators, and let the output reference generator have nondegenerate packet energies $(n_0 + r)\Delta$, $\Delta > 0$. Its reduced reference ω_R then obeys

$$D_{\text{tr}}(\omega_R, |\eta_L\rangle\langle\eta_L|) \geq 1 - \frac{1}{L}.$$

Proof. Covariance preserves total stationarity. Partial trace under an additive output generator leaves a stationary reference. Its matrix is diagonal in the nondegenerate energy basis, so $\langle\eta_L|\omega_R|\eta_L\rangle = L^{-1} \sum_{r=0}^{L-1} (\omega_R)_{n_0+r, n_0+r} \leq 1/L$. The two-outcome measurement of the target projector gives the distance lower bound. The uniform diagonal packet has eigenvalues $1/L$ on the packet block; its difference from the pure uniform packet has trace norm $2(1 - 1/L)$, attaining the bound. $\square$

Any coherent fuel, reference, interacting controller or spectator that breaks complete stationarity must be counted. The theorem does not forbid preparation from admitted nonstationary resources, nor does it give a universal approximate-QFI cost. Degeneracy across packet energies would invalidate the diagonal step and requires a different statement.

4. Supplied history word and finite clock

Supply gates $U_1, \dots, U_n$ commuting with a comparison work Hamiltonian $\bar{H}_W$. Define prefixes $P_0 = I$, $P_j = U_j \dots U_1$, and

$$\mathcal{W} = \sum_{j=0}^n |j\rangle\langle j|_C \otimes P_j, \quad H_C = \kappa \sum_{j=0}^{n-1} \sqrt{(j+1)(n-j)} (|j+1\rangle\langle j| + |j\rangle\langle j+1|),$$

with $\kappa > 0$. Let $H_p = \mathcal{W}(H_C \otimes I)\mathcal{W}^\dagger$ and

$$H_{\text{act}} = I_C \otimes H_W + H_p, \quad H_{\text{comp}} = I_C \otimes \bar{H}_W + H_p, \quad \|H_W - \bar{H}_W\| \leq d.$$

Gate-dependent hopping edges contain $U_j, U_j^\dagger$. Their synthesis is supplied; a nearest-neighbor address graph does not establish locality on the work registers.

Theorem 4 (history propagation). The clock has spectrum $\kappa(n - 2j)$, $j = 0, \dots, n$, norm κn and amplitudes

$$c_j(t) = (-i)^j \sqrt{\binom{n}{j}} \sin^j(\kappa t) \cos^{n-j}(\kappa t).$$

From address 0, the comparator applies P_n at $T = \pi/(2\kappa)$ with terminal probability one, and returns to prefix P_0 at $2T$. Actual/comparator propagator difference is at most td .

Proof. Identify address j with the normalized symmetric sum of n binary spin states of weight j . The operator $\kappa \sum_{a=1}^n X_a$ has the displayed adjacent matrix elements and restricts to H_C . Tensor products of single-spin rotations give $c_j(t)$; symmetric products of X eigenstates give the spectrum. Since $[\bar{H}_W, \mathcal{W}] = 0$,

$$e^{-itH_{\text{comp}}}|0\rangle|\psi\rangle = \sum_j c_j(t)|j\rangle e^{-it\bar{H}_W} P_j |\psi\rangle.$$

At T only $j = n$ survives; at $2T$ only $j = 0$ survives. The actual difference is $H_W - \bar{H}_W$, so Lemma 1 gives the last bound. The identity extends to mixed inputs and arbitrary spectators by purification. $\square$

The endpoint return is projective: the common factor at $2T$ is $(-1)^n$. Free work evolution is retained; the full work state need not recur. At integer endpoint index $\ell = 2u + r$, $\ell \equiv r \pmod{2}$, the address is start for $r = 0$ and terminal for $r = 1$. Returning the address does not refill a donor or provide a stopping rule.

Corollary 5 (completion window). Start the actual machine at address 0 with arbitrary work/spectator state. For $t = T + s \geq 0$ and $|s| \leq \sigma$, define

$$a_C = \sqrt{n} \kappa \sigma + (T + \sigma)d.$$

Its nonterminal probability is at most $\min\{1, a_C^2\}$.

Proof. The comparison terminal probability is $\cos^{2n}(\kappa s)$. The inequality $1 - (1 - z)^n \leq nz$ for $z \in [0, 1]$, together with $|\sin(\kappa s)| \leq \kappa|s|$, bounds the nonterminal amplitude by $\sqrt{n} \kappa \sigma$. Lemma 1 adds at most $td \leq (T + \sigma)d$ to this projected amplitude. $\square$

If the source terminal-branch target has its inherited isolated bank error η_{bank} and the implemented/comparison post-endpoint free phases differ by at most $|s|d$, a complete target bound is

$$\eta_{\text{window}} \leq \min\{1, \eta_{\text{bank}} + \sqrt{n} \kappa \sigma + (T + 2\sigma)d\}.$$

Here the comparator is compared to its normalized terminal branch, whose purification overlap has magnitude $|c_n|$, hence trace distance $\sqrt{1 - |c_n|^2}$. Triangle inequality then supplies the displayed bound. The branch-target premise is essential: a generic arbitrary word is not a packet producer. The whole propagation mismatch replaces the earlier separately timed pulse debit; both are not charged for the same comparison interval.

5. Timed terminal capture and protected propagation

5.1 Guarded background and matched involution

Supply matched S, O with $H_O = H_S$ and fresh $L = 0$. Records M, E are bare-degenerate. The production clock and flag use zero bare park energies in this architecture. Define

$$H_{\text{free}} = H_W + H_O, \quad H_{\text{bg}} = H_{\text{free}} + P_{L0} H_p, \quad P_{L0} = |0\rangle\langle 0|_L.$$

The guard commutes with the flag and has norm at most κn . O has no background interaction: $H_{\text{bg}} = H_O + H_{\text{rest}}$. This conditional guard is an engineered input.

With terminal projector P_n on C , set

$$J = P_n \otimes \text{SWAP}_{SO} \otimes X_L + (I - P_n) \otimes I, \quad G_c = \frac{\pi}{2\tau_c}(I - J), \quad \tau_c > 0.$$

Matched bare energies imply $[J, H_{\text{free}}] = 0$, and $J^\dagger = J$, $J^2 = I$. Thus $e^{-i\tau_c G_c} = J$ and $\|G_c\| \leq \pi/\tau_c$, with equality when a negative-eigenvalue sector is present. The pulse swaps unknown states and correlations; it does not clone them or create a blank output.

Theorem 6 (timed contact and flags). For comparison $B_c = e^{-i\tau_c H_{\text{free}}} J$ and actual $A_c = e^{-i\tau_c (H_{\text{bg}} + G_c)}$,

$$\|A_c - B_c\| \leq \gamma_c = \tau_c \kappa n.$$

For an incoming state with fresh flag and nonterminal probability at most a_C^2 ,

$$p_{\text{NO_CAPTURE}} \leq \min\{1, (a_C + \gamma_c)^2\}, \quad p_{\text{false tag}} \leq \min\{1, \gamma_c^2\}.$$

The false terminal-tag event is $(I - P_n)P_{L1}$, not every flag-1 outcome.

Proof. The ideal pulse commutes with the free generator; actual/comparison generator difference is the guarded hopping of norm at most κn . Lemma 1 proves the uniform contact bound. Ideal capture sends terminal fresh-flag inputs to flag 1 and nonterminal inputs to flag 0, so its no-capture amplitude equals the incoming nonterminal amplitude. The bad event annihilates ideal capture on every fresh-flag input because its flag-1 branch has terminal C . Projected amplitude triangle inequalities give both bounds. Free evolution preserves these projectors. $\square$

Additional active-contact integrated norms enter γ_c only when their domains match. Initial supply error changes probabilities by at most the admitted joint distance. False-tag probability is a diagnostic already controlled by the unconditional joint bound; it is not charged again as an independent state error.

5.2 Full record, witness and pulse-off theorem

Copy the commuting flag to blank M , then to blank E using controlled X . These copy unitaries commute with H_{bg} ; a Hermitian logarithm for each can also be chosen to commute, so ideal simultaneous background evolution factors. Supply and implementation of the copy controls remain costs. The complete dilation retains L, M, E and all correlations. After reducing witness E , the operational record is

$$\mathcal{R}(\rho) = \sum_{\ell=0,1} |\ell\rangle\langle\ell|_M \otimes P_{L\ell} \rho P_{L\ell}.$$

Both branches are unnormalized. A classical operational reduction does not make the full pure dilation a classical state.

Theorem 7 (post-switch output protection). After the supplied capture and record contacts are off, evolution under H_{bg} gives for any actual input, including failures and false tags,

$$\rho_{OM}(u) = (e^{-iuH_O} \otimes I_M) \rho_{OM}(0) (e^{iuH_O} \otimes I_M).$$

Flag probabilities remain constant. In flag 1 the production hopping is parked.

Proof. The background factors as $H_O + H_{\text{rest}}$ and acts trivially on M . In taking the reduced OM state, conjugation by the rest unitary disappears under partial trace, even for correlated inputs. Commutation with $P_{L\ell}$ preserves flag populations; $P_{L0}H_p$ vanishes in flag 1. $\square$

Protection is exact for the isolated declared background, not a measured lifetime. A coupling to O or a flag-flipping perturbation needs its own integrated bound. Common implemented free output phases remain in the target; no new comparator mismatch growing with storage time is charged when actual and target share that evolution.

Lemma 8 (restricted capture posterior). If the operational classical-quantum state has distance ϵ from an ideal state supported wholly on CAPTURED with target σ , and its actual capture probability is $p > 0$, then its normalized captured state has distance at most $\min\{1, \epsilon/p\}$ from σ .

Proof. The block-diagonal norm gives $\epsilon \geq \frac{1}{2}\|p\rho_{\text{cap}} - \sigma\|_1 + (1-p)/2$. Since $p(\rho_{\text{cap}} - \sigma) = p\rho_{\text{cap}} - \sigma + (1-p)\sigma$, triangle inequality gives $pD_{\text{tr}}(\rho_{\text{cap}}, \sigma) \leq \epsilon$. $\square$

This lemma does not apply unchanged to arbitrary detector outcomes whose ideal acceptance is below one. Capture/export timing, pulse-off and routing are supplied controls.

6. Closed finite first-entry obstruction

Theorem 9 . For one finite time-independent Hermitian H , a subspace invariant under H is reducing. Its population cannot first increase from the orthogonal complement. If a fixed observable expectation under e^{-itH} is constant on an open time interval, it is constant for all real t .

Proof. Hermiticity makes $\langle Hx, y \rangle = \langle x, Hy \rangle = 0$ for x in the orthogonal complement and y in the invariant subspace. Its projector therefore commutes with H and with the propagator; population is constant. Spectral decomposition expresses any expectation as a finite sum $\sum_{a,b} c_{ab} e^{it(E_a - E_b)}$, a real-analytic function of real time. A constant on an open interval has identically zero derivative by analytic continuation. $\square$

This is a restricted obstruction to exact permanent first-entry capture in a closed finite static model. It excludes neither finite-horizon approximate retention, a supplied change of Hamiltonian, nor an open reservoir with its own ledger. If the ideal G_c stays on for $2\tau_c$, $J^2 = I$ undoes the exchange. With a noncommuting background, exact reversal is not asserted; its deviation is a perturbation comparison.

7. Static post-launch capture and finite horizon

7.1 Address dressing

Retain $H_f = H_W + H_O$ and parent guarded background $H_b = H_f + V_p$, $\|V_p\| \leq h_p$. Supply fresh $L, M, E = 0$, terminal production projector P_n , and

$$J_* = P_n \otimes \text{SWAP}_{SO} \otimes X_L X_M X_E + (I - P_n) \otimes I.$$

Then $J_*^2 = I$ and $[J_*, H_f] = 0$. The phase conventions, matched energies and blank records are inputs.

Supply clock Q initially at address 0 and define H_Q by the spin-clock formula of Section 4 with n, κ replaced by N, λ , $N \geq 1$, $\lambda > 0$. Put $W_0 = I$, $W_j = J_*$ for $j \geq 1$ and

$$\mathcal{V} = \sum_{j=0}^N |j\rangle\langle j|_Q \otimes W_j, \quad H_d = \mathcal{V}(H_Q \otimes I)\mathcal{V}^\dagger, \quad H_a = I_Q \otimes H_b + H_d, \quad H_0 = I_Q \otimes H_f + H_d.$$

Only edge 0-1 carries J_* ; subsequent edges carry identity. The dressed generator has norm λN . All couplings are constant after a supplied launch; launch alignment is not derived.

7.2 Complete versus marginal comparison

Theorem 10 (static joint bound). Set $r(t) = \cos^{2N}(\lambda t)$ and let $|\chi(t)\rangle = e^{-itH_Q}|0\rangle$. For any body/spectator state ρ , the actual evolution from $Q = 0$ satisfies

$$D_{\text{tr}}(\rho_a(t), |\chi(t)\rangle\langle\chi(t)| \otimes e^{-itH_f} J_* \rho J_* e^{itH_f}) \leq \min\{1, th_p + 2\sqrt{r(t)}\}.$$

In the comparison alone, reducing Q produces a mixture with weight r on the pre-capture body and $1-r$ on the transferred body, so its reduced-body distance from the latter is at most r .

Proof. Section 4 and $[J_*, H_f] = 0$ give on a purification

$$|\Psi_0(t)\rangle = \sum_{j=0}^N c_j(t) |j\rangle e^{-itH_f} W_j |\psi\rangle.$$

Against the displayed fully transferred product target, the vector difference is $c_0(t)|0\rangle e^{-itH_f}(I - J_*)|\psi\rangle$, with norm at most $2|c_0| = 2\sqrt{r}$. Actual/comparison propagator difference is bounded by th_p . Lemma 1 and triangle

inequality give the complete bound. Orthogonal address reduction gives exactly the stated mixture; convexity bounds its distance by r . $\square$

The marginal r cannot replace the complete $2\sqrt{r}$ when retaining the clock or reusing correlated resources. The target clock continues to evolve; it is not a classical scheduler.

For fresh records, incoming production nonterminal probability p gives comparison NO_CAPTURE probability $p + (1-p)r$. If $\sqrt{p} \leq a$, its amplitude is at most $a + \sqrt{r}$ and the actual probability is at most

$$\min\{1, (a + \sqrt{r} + th_p)^2\}.$$

The false-tag projector $(I - P_n)P_{L1}$ annihilates the comparison on fresh records, so its actual probability is at most $\min\{1, (th_p)^2\}$. As before, initial supply and extra contacts require additional admission; diagnostic probabilities do not become independent joint-error charges.

7.3 Window, return and resource tradeoff

Corollary 11 (finite retention interval). On

$$\frac{\pi}{4\lambda} \leq t \leq \frac{3\pi}{4\lambda}, \quad r(t) \leq 2^{-N},$$

the complete debit is at most $\min\{1, 3\pi h_p/(4\lambda) + 2 \cdot 2^{-N/2}\}$. Comparator transfer is exact at $\pi/(2\lambda)$ and is undone at π/λ .

Proof. On the interval $|\cos(\lambda t)| \leq 1/\sqrt{2}$, and $t \leq 3\pi/(4\lambda)$. Theorem 10 gives the bound. At half a spin period the clock is at N , dressed by J_* ; at a full projective period it is at 0, dressed by identity. $\square$

The last return includes record flips and free body evolution; it is not storage permanence. Actual departure from comparison is charged, so exact comparison recurrence is not an exact actual recurrence theorem. These are pointwise bounds, not continuously monitored first-entry or survival probabilities. Monitoring changes the channel.

Write interval duration $\mathcal{H} = \pi/(2\lambda)$. Its conservative parent-contact term is $3\mathcal{H}h_p/2$, while clock norm is $\pi N/(2\mathcal{H})$. Padding decreases the boundary tail but not that parent term. This is a sufficient-bound tradeoff, not a lower bound on every possible apparatus.

For a fixed allowed static-clock spectral norm $g_Q = \lambda N$, substitution yields

$$\mathcal{H} = \frac{\pi N}{2g_Q}, \quad \epsilon_{\text{static}} \leq \min\left\{1, \frac{3\pi h_p N}{4g_Q} + 2^{1-N/2}\right\}.$$

Thus the conservative complete-state bound need not improve monotonically with N at fixed g_Q . This is an immediate consequence of the stated construction, not a universal limit on clocks. Likewise a history clock with fixed norm $g_C = \kappa n$ has ideal completion time $T = \pi n/(2g_C)$. Long retention requires another owned shutdown, routing or reservoir mechanism.

8. Matched renewal and finite resource banks

8.1 Exact correlation export

Theorem 12 (matched renewal). For a finite active module A , an independent matched donor D in target state σ_D , and arbitrary correlated old state ρ_{AY} ,

$$\text{SWAP}_{AD}(\rho_{AY} \otimes \sigma_D)\text{SWAP}_{AD} = \sigma_A \otimes \rho_{DY}.$$

Every old A/Y correlation is transferred to D/Y .

Proof. Expand ρ_{AY} in matrix units on A and Y and σ_D in matrix units on D . Conjugating by SWAP exchanges the two identified module labels. Reordering tensor factors gives the displayed product. No partial trace or success conditioning is used. $\square$

For timed capture, $A = (S, R, F, C, L)$ while outputs, records and witnesses remain in Y . For static capture, include Q in A . A replacement donor must include fresh coherent fuel, loaded reference, prepared clock and flag, any needed fresh source cells, and the appropriate endpoint free phase. Merely restoring a clock address does not restore the whole module. Exact independence of the ideal donor is a hypothesis; approximate correlations enter complete bank error.

For duration τ_r , the matched principal-log exchange generator is $G_r = \pi(I - \text{SWAP})/(2\tau_r)$. It commutes with the matched sum of bare free operators. Actual running interactions add architecture-specific full-joint contact bounds:

Architecture	Running operator charged	Contact upper bound
Original dressed controller against bare clock	$H_p - H_C$, norm at most $2\kappa n$	$2\tau_r \kappa n$
Guarded timed capture	$P_{L0} H_p$, norm at most κn	$\tau_r \kappa n$
Static capture including Q	Guarded parent plus H_d , norm at most $h_p + \lambda N$	$\tau_r (h_p + \lambda N)$

The comparator/free split determines the debit. Copying a timed norm into static renewal would omit its capture clock. Static full-module renewal in the source is an algebraic continuation; no implemented autonomous renewal/router is certified.

8.2 Whole-bank admission and unconditional composition

Proposition 13 (finite sequence budget). Let the complete supplied resource bank B and history Y satisfy

$$D_{\text{tr}}(\omega_{BY}, \sigma_B \otimes \omega_Y) \leq \epsilon_F,$$

where the ideal bank specifies independent fresh slots and all blanks. Suppose each actual operation differs from its matching ideal operation by at most ϵ_j on the full admitted joint state, with arbitrary spectator extension and the same retained domain. After A operations,

$$D_{\text{tr}}(\rho_A, \rho_A^{\text{ideal}}) \leq \min \left\{ 1, \epsilon_F + \sum_{j=1}^A \epsilon_j \right\}.$$

Proof. For actual channel Φ_j and ideal Ψ_j , insert the ideal input at step j . Contractivity bounds the first difference by the preceding joint error; the assumed operation comparison bounds the second by ϵ_j . Induction starts with ϵ_F . If different operations expose different registers, extend both by the same identity/free channel on all retained registers before using this argument. $\square$

A bound proved only for an isolated input does not automatically satisfy this uniform-domain hypothesis. The per-operation comparisons may be certified uniformly by a channel norm, or directly for every input in the admitted joint-state domain. Interleaving contacts on the same reference require a complete integrated comparison. Production, capture, renewal, routing, detector, storage, thermal and implementation terms are summed only on matching contracts. No-capture and false-tag diagnostics already covered by state distance are not double-counted. Full-bank admission is charged once; consumed slots and all failed histories remain.

Good marginals need not imply a good bank. Label bit 1 a successful slot. The uniform classical distribution on 110, 101, 011 has marginal success $2/3$ at each of three slots but no 111 outcome. Its complete distance from the all-success target is one, whereas the unjustified independent product would give success $8/27$. This witness rules out a marginal-product admission, not a valid complete correlated-state admission.

For A supplied donor and output slots, attempts are $j = 0, \dots, A - 1$; at $j = A$ the specified controller refuses and appends END_OF_SUPPLY. There is no modular wrap. Failure consumes a slot as success does. Reusing an old output as blank can swap previous data back into the machine. Slot selection, refusal and routing remain supplied policy, not a derived autonomous counter. Donor counts are finite supplied inputs.

9. Energy conventions, records and implementation debts

During a constant machine interval total mean energy is conserved. For $H_{\text{act}} = H_W + H_C + V$, $V = H_p - H_C$,

$$\Delta\langle H_{\text{act}} \rangle = 0, \quad \Delta\langle H_W + H_C \rangle = -\Delta\langle V \rangle, \quad \|V\| \leq 2\kappa n.$$

This does not assert conservation of every subsystem's bare energy. Adding one global constant alters an energy origin without changing trajectories. A flag-conditioned offset changes the Hamiltonian and cannot be inserted as an innocent convention.

For the supplied constant capture pulse, switch-on/off signed mean work is

$$W_c = \langle G_c \rangle_{\text{in}} - \langle G_c \rangle_{\text{out}} = \Delta\langle H_{\text{bg}} \rangle, \quad |W_c| \leq 2\pi\kappa n.$$

Indeed $[H_{\text{free}}, G_c] = 0$, so the rate of $\langle G_c \rangle$ is governed by the guard; integrating $|\langle i[P_{L0}H_p, G_c] \rangle| \leq 2\kappa n\|G_c\|$ over τ_c gives the bound. It is a signed mean-work estimate, not a work distribution, peak-power certificate or supply mechanism. An ideal commuting flag-copy pulse has zero net signed mean switching work under that convention, but consumes blank memory and creates correlations; record erasure is not free.

For fresh static clock address 0, $\langle H_d \rangle = 0$ and $\langle H_d^2 \rangle = \lambda^2 N$ for every body state. The first statement follows from off-diagonal address support; the second from the first edge's squared magnitude $\lambda^2 N$ and $J_*^2 = I$. Body-only background cross covariance vanishes initially, so $\text{Var } H_a = \text{Var } H_b + \lambda^2 N$. A global shift by λN changes the mean but not the variance. Address 0 is nonstationary under the engineered generator; zero mean in one convention does not imply free clock preparation.

The following ledger must accompany any finite example.

Resource	Required specification	Unclosed acquisition/implementation gate
Production clock	Dimension, state, norm, jitter, recurrence	Prepared supply, edge synthesis and launch
Reference/fuel	Complete state, seed energy, variance, shift margins	Coherent source and energy hierarchy
Timed capture	Matched carriers, duration, pulse and off-switch	Trigger, control tolerance and switching apparatus
Static capture	Fresh clock, full operator, horizon and return	Alignment, local coupling and export
Donor bank	Whole-bank error, spectator correlations, finite cap	Independent donors, routing and refusal
Output/records	Fresh slots, retained witness, acceptance convention	Calibration, acquisition and storage perturbations

These specifications distinguish a prepared operator construction from a complete autonomous cycle. They must be included in any proposed physical implementation.

10. Exact examples, replay and limitations

Take the declared example $n = 23$, $\kappa = 4096 = 2^{12}$, so $h_p = 94208 = 23 \cdot 2^{12}$. With supplied timed contact $\tau_c = 2^{-30}$,

$$\gamma_c = \frac{23}{2^{18}}, \quad \gamma_c^2 = \frac{529}{2^{36}}.$$

For static $N = 40 = 2^3 \cdot 5$, $\lambda = 2^{32}$, clock dimension is 41 and norm $\lambda N = 5 \cdot 2^{35}$. The uniform interval clock tail is

$$r \leq \frac{1}{2^{40}}, \quad 2\sqrt{r} \leq \frac{1}{2^{19}}, \quad \frac{3\pi h_p}{4\lambda} = \frac{69\pi}{2^{22}}.$$

The interval begins at $\pi/2^{34}$ and ends at $3\pi/2^{34}$ after supplied launch. Its duration is $\pi/2^{33}$ in model units. Static renewal at $\tau_r = 2^{-50}$ has contact upper bound

$$\frac{h_p + \lambda N}{2^{50}} = \frac{41943063}{274877906944}.$$

These values compare different declared architectures; their error columns are not interchangeable or summed as one apparatus. Huge dimensionless norms, very short horizons, coherent clock supply and unknown laboratory conversion are implementation debts. The parameters are supplied example inputs, not derived coefficients selected to match an observation.

The executable `replay/verify_paper_iv.py` supplies exact finite-map checks and separate numerical joint comparisons. It includes spectator-entangled inputs, clock transfer/return, the capture involution, matched-energy commutation, static full-joint versus marginal distance, renewal correlation export, stationary reference bound and a correlated-bank negative control. It does not instantiate the large source parameter example, certify all acquisition conditions or establish the general proofs by enumeration. Its result receipt records actual executed counts and tolerances. The M4 check set and the original baseline are separate receipts.

The 5 October 2026 ingestion replay reproduces 378 exact controls and 71 numerical fixtures, with numerical tolerance 2×10^{-10} and maximum recorded residual about 1.30×10^{-14} . The original chain baseline and M4 algebra suite separately reproduce 1,119 and 116 exact checks. These are constructor replays with shared lineage; their counts are implementation checks rather than an independent proof review or an empirical result.

The static clock's $N = 40$ denotes 41 address states, not 40 address qubits. A binary encoding of the address alone requires at least $\lceil \log_2 41 \rceil = 6$ qubits; a single-excitation encoding may use 41 physical two-level sites. Neither representation includes all work, fuel, reference, output, witness, and donor registers. Synthesis cost depends on the selected encoding. The times above have no asserted laboratory conversion, so a physical duration cannot be inferred from their small numerical model-unit values.

11. Engineering implications and remaining questions

The bounds provide design constraints after resource and control specifications are supplied. For example, increasing static-clock size suppresses the transfer-tail term, but the perturbation term depends on the clock rate and the desired retention interval. At a fixed rate it does not disappear merely by adding addresses. At a fixed spectral-norm budget, increasing addresses also lengthens the interval and can increase the accumulated parent-interaction bound. These dependencies should be evaluated jointly before an encoding or physical platform is selected.

A publication-strength computational study should sweep word length, jitter, mismatch, retention duration, and donor-bank correlations. It should compare exact simulated joint error with the analytic upper bound, report saturation and loose regimes, and retain failure branches. A reduced-body baseline should be plotted separately to illustrate precisely when it is inadequate for reuse. Any comparison with other finite clocks must align initial resources, interaction strength, preparation assumptions, observable targets, and error metrics; the present draft establishes no such superiority result.

A bounded production-to-capture trigger and finite router on a single retained joint domain remain development targets. The architecture does not derive launch alignment, coherent resource preparation, pulse-off control, fresh-slot selection, or exhaustion refusal. A finite refusal policy is stated in Section 8, but a physical autonomous counter is not constructed. Coupling locality and hardware calibration require additional implementation models.

The principal result is a conditional accounting framework for finite quantum execution, capture, and renewal. It preserves complete-state errors, exported correlations, switching-work conventions, and resource exhaustion in one analysis. Energy-preserving exchanges do not imply zero dissipation or free preparation. Correlation transfer does not imply an indefinitely reusable catalyst. The framework supplies neither a laboratory realization nor a fault-tolerance threshold.

Data and code availability

The accompanying development package includes the four unmodified M5 manuscript sources, their hashes, the original replay scripts and data, and the separately executed ingestion receipts. The scripts require Python; the controller replay additionally requires NumPy and SciPy. Small fixtures implement specified finite instances and negative controls. They do not instantiate the large example’s complete apparatus or verify all acquisition assumptions. Original nested provenance archives remain in the source release identified in the ingestion record. This draft and its computational checks are not represented as independently reviewed publications.

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